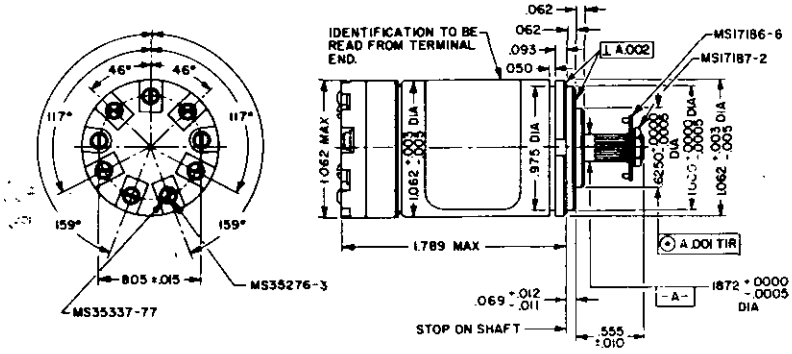
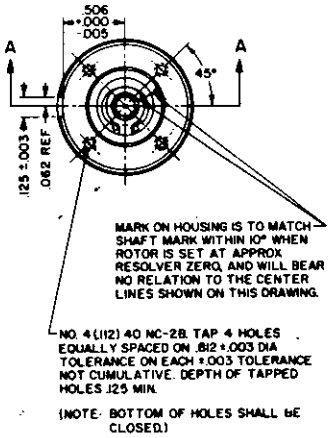
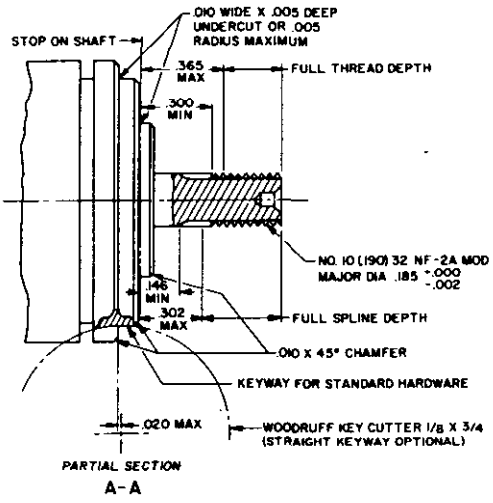
# MILITARY SPECIFICATION SHEET
# RESOLVER TRANSMITTER, TYPE .25V-11R2N2a

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of MIL-R-23417.




| SPLINE DATA | |
|---|---|
| TEETH | 21 |
| PITCH | 120 |
| PITCH DIAMETER | .1750 +.0000 -.0020 |
| OUTSIDE DIAMETER | .1872 +.0000 -.0005 |
| PRESSURE ANGLE | 20° |
| ROOT DIAMETER MAXIMUM | .155 |
| TOOTH FORM | FULL DEPTH INVOLUTE |
| NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. SPLINE SHALL BE INSPECTED BY GO, NO-GO SPLINE RING GAGE. MS90406-1. | |

NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING ROTATED.
2. RUNOUT OF FEATURE -A- (SMOOTH PORTION) OF SHAFT SHALL NOT EXCEED .0008 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS
   END PLAY - 1 POUND
   RADIAL PLAY - 1/2 POUND
4. THE END CAP AND TERMINAL BLOCK SHALL BE SECURED BY SCREWS OTHER THAN THE TERMINAL SCREWS. RECESSES SHALL ALLOW TERMINAL LUGS *MS17182-2 TO BE INSTALLED WITH THE BARREL IN-*BOARD OR OUTBOARD AND PREVENT THE LUGS FROM TURNING.
5. THE FOLLOWING LIST OF HARDWARE MS35276-3 MACHINE SCREW (7 REQUIRED), MS35337-77 LOCK WASHER (7 REQUIRED), MS17182-2 TERMINAL LUG (7 REQUIRED), MS17186-6 DRIVE WASHER (1 REQUIRED), MS17187-2 NUT (1 REQUIRED), SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS, THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT RESOLVER PACKAGE.
6. COLLECTOR RING MATERIAL SHALL BE 69 GOLD, 06 PLATINUM AND 25 SILVER.
7. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE RESOLVERS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN MIL-R-23417.



FSC 5990

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TABLE I

RESOLVER TRANSMITTER, TYPE .25V-11R2N2a
PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

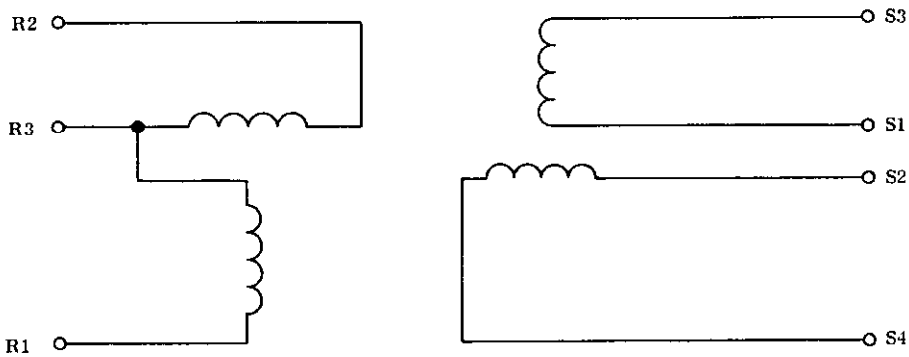
| REQUIREMENTS | VALUE | UNIT | TOLERANCE |
|---|---|---|---|
| Rated frequency | 135 | cps | ±1% |
| Input voltage | 0.25 | Volts | ±10% |
| Test voltage | 0.25 | Volts | ±1% |
| Input current | 3.0 | Milliamps | Nominal |
| Input power | 0.00041 | Watts | Nominal |
| Angular error | 7.0 | Minutes | Maximum |
| Transformation ratio | 3.1 | ---- | ±4% |
| Null voltage: | | | |
| Total | 4.0 | Millivolts | Maximum |
| Fundamental | 3.0 | Millivolts | Maximum |
| Temperature rise | 20.0 | Degrees (C) | Maximum |
| Phase shift (Input to output secondary open circuit) | 28.0 | Degrees | ±4.0 |
| Friction torque: | | | |
| Room temperature | 10.0 | Gm-Cm | Maximum |
| -54 to 85° C | 20.0 | Gm-Cm | Maximum |
| Impedance: | | | |
| Zro | 90-110 | Ohms | Min.-Max. |
| Zss | 940-1150 | Ohms | Min.-Max. |
| Impedance angle: | | | |
| Zro | 57.0-63.0 | Degrees | Min.-Max. |
| Zss | 16.0-21.0 | Degrees | Min.-Max. |

Notes:

1. The Resolver Transmitter identified herein shall conform to all requirements and shall be subjected to all tests delineated in MIL-R-23417.

2. The type .25V-11R2N2a Resolver was formerly known as the .25V-11RX2a, also as the 11R16N2 Mk 7 Mod O.

3. Sampling, inspection and preparation for delivery shall be in accordance with MIL-R-23417.




.25V/.775V
CONNECTION DIAGRAM



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Custodian:

Navy-AS

Preparing activity:

Navy-As

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